

Selva Ganesh V

9600964798 | selvaganesh0809@gmail.com | linkedin.com/in/selva-ganesh-v-330411237 | github.com/SelvaGaneshV

PROFESSIONAL SUMMARY

Software Engineer with 2+ years of experience building scalable full-stack applications using **React, TypeScript, and NestJS**, specializing in **multi-tenant systems**, complex UI editors, and workflow platforms with concurrency control.

EDUCATION

Francis Xavier's Engineering College

Master of Computer Applications

Tirunelveli, TN, IN

Oct. 2021 – June 2023

St.John's College

Bachelor of Computer Applications

Tirunelveli, TN, IN

June 2018 – April 2021

EXPERIENCE

Software Engineer

Torus Innovations

Oct 2023 – Present

Tirunelveli, TN, IN

- Designed and developed a **node-based visual workflow editor** using React, TypeScript, and React Flow, supporting **thousands of workflow artifacts** and adopted by **35+ internal users**.
- Built a grid-based **WYSIWYG UI editor** with drag-and-drop positioning, resizing, and alignment helper lines; migrated **2 internal applications (10+ screens)** from manual layout configuration.
- Implemented **artifact-level locking using Socket.IO**, eliminating **100% of daily concurrent modification conflicts** across **20+ actively edited artifacts** in multi-user environments.
- Architected a **schema-driven, runtime-extensible node configuration system** enabling admin-defined node types via JSON upload; new node types become immediately available in the workflow editor without front-end redeployment.
- Engineered **35+ RESTful APIs (NestJS)** powering workflow graph normalization and execution readiness, including circular dependency detection, cross-route referential validation, nested graph mutations, and standardized payload generation for downstream code-generation services.
- Contributed to **multi-tenant SaaS** architecture with Role-Based Access Control (RBAC), ensuring secure data isolation across tenants.

PROJECTS

VezaI.db | React, TanStack Start, AI SDK, Hono, PostgreSQL, Drizzle, Turborepo

Aug 2025 – Present

- Designed and implemented a scalable **monorepo architecture using TurboRepo**, with shared modules for authentication, database abstraction, and caching.
- Developed a full-featured **database management interface** including ER diagram visualization, schema editor, and table data editor.
- Implemented secure **authentication and authorization** using BetterAuth and JWT for multi-user access control.
- Built an **AI-powered database assistant** capable of understanding relational and NoSQL schemas and generating context-aware queries using an AI SDK.
- Optimized application scalability and modularity, enabling seamless support for additional database engines and feature extensions.

TECHNICAL SKILLS

Programming Languages: JavaScript (ES6+), TypeScript

Frontend: React, Next.js, Svelte, React Flow, TanStack Start, Vite

Backend: Node.js, NestJS, REST APIs, Hono

Databases: PostgreSQL

Caching: Redis, LRU Cache

Real-Time Communication: Socket.IO

Authentication & Security: JWT, RBAC, BetterAuth

Architecture & Tools: Multi-tenant SaaS, Monorepo (TurboRepo), Git